



Benchmarking Text Classification Models

History of Language Models

2017

Transformers
Created:
Attention is all
you need

2018

BERT: Pre-training of
deep bidirectional
transformers for
language
understanding

2021

Want to reduce
labeling cost? GPT-3
can help

2021

GLaM: Efficient Scaling
of Language Models
with Mixture-of-
Experts

Late 2021

Want to Reduce
Labeling: Cost? GPT-3
Can Help

Our Goal

Our mission is to explore how **few-shot learning** and **active learning** methods can boost the accuracy of Large Language Models (LLMs) such as GPT, BERT, and SetFit.

In the realm of AI/ML, where obtaining ample labeled data is challenging, especially for unstructured text data, we focus on using a continuous human feedback loop that involves human input. This process aims to fine-tune models and enhance model accuracy, recall, and precision using a limited number of labeled examples.

The Results

Our results demonstrate consistent improvement in accuracy across diverse text datasets, showcasing the potential of few-shot learning and active learning to refine language models efficiently with minimal labeled data and human feedback.

Dataset

Amazon Reviews

Product Reviews

Excellent

Very Good

Neutral

Good

Bad

Financial Phrasebank

Financial phrases

Positive

Negative

Neutral

sentence string · lengths 	label class label  3 classes
According to Gran , the company has no plans to move all production to Russia , although that is where the company is growing .	1 neutral
Technopolis plans to develop in stages an area of no less than 100,000 square meters in order to host companies working in computer technologies and telecommunications , the statement said .	1 neutral
The international electronic industry company Elcoteq has laid off tens of employees from its Tallinn facility ; contrary to earlier layoffs the company contracted the ranks of its office workers , the daily Postimees reported .	0 negative
With the new production plant the company would increase its capacity to meet the expected increase in demand and would improve the use of raw materials and therefore increase the production profitability .	2 positive
According to the company 's updated strategy for the years 2009-2012 , Basware targets a long-term net sales growth in the range of 20 % -40 % with an operating profit margin of 10 % -20 % of net sales .	2 positive

Self Learning Process

The Market

Zero-shot Models for Spam Prediction

Leveraging Claude, GPT3.5, BERT, and SETFIT in text classification.

Initiation of predictions involves identifying area where model prediction **incorrectly predicted**.

Table 1: Zero shot model predictions

Text Body	Predicted	Probability	Entropy
Important notice: Your package has been delivered.	not spam	0.65	0.88
Dear customer, Your account balance is low.	not spam	0.58	0.82
Hi, How are you doing? Let's catch up soon.	not spam	0.58	0.75
Urgent notice: Last chance to update your personal information.	spam	0.92	0.51
Hi there, You have won a free vacation! Claim now!	spam	0.95	0.42
Congratulations! You've won a million dollars!	spam	0.97	0.36

Our Solution

Iterative Refinement with Human Feedback

By **continuously refining human feedback** early on, providing the actual answer in row 1 of Table 3, we address initially incorrect predictions, such as in cases like “Dear customer, Your account balance is low,” where our model previously predicted 'not spam,' but now the model learn to predict that the actual answer is 'spam.'

Table 3: Iterated Model Predictions - First Iteration

Text	Actual Label	Predicted	Probability	Entropy
Important notice: Your package has been delivered.	not spam	not spam	1.0	0
Dear customer, Your account balance is low.		spam	0.70	0.76
Hi, How are you doing? Let's catch up soon.		not spam	0.65	0.68
Urgent notice: Last chance to update your personal information.		not spam	0.88	0.28
Hi there, You have won a free vacation! Claim now!		spam	0.80	0.25
Congratulations! You've won a million dollars!		spam	0.92	0.22

Fine-Tuning

The Market

Zero-shot Models for Spam Prediction

With GPT 3.5, it sometimes generates inaccurate predictions or explanations even when a simple format is preferred

The statement "the market is up today" generally implies a positive movement in the financial markets. When people say that the market is up, they usually mean that stock prices, as represented by market indices, have increased. This is often seen as a positive sign, indicating overall confidence in the economy and the companies listed on the stock market.

Our Solution

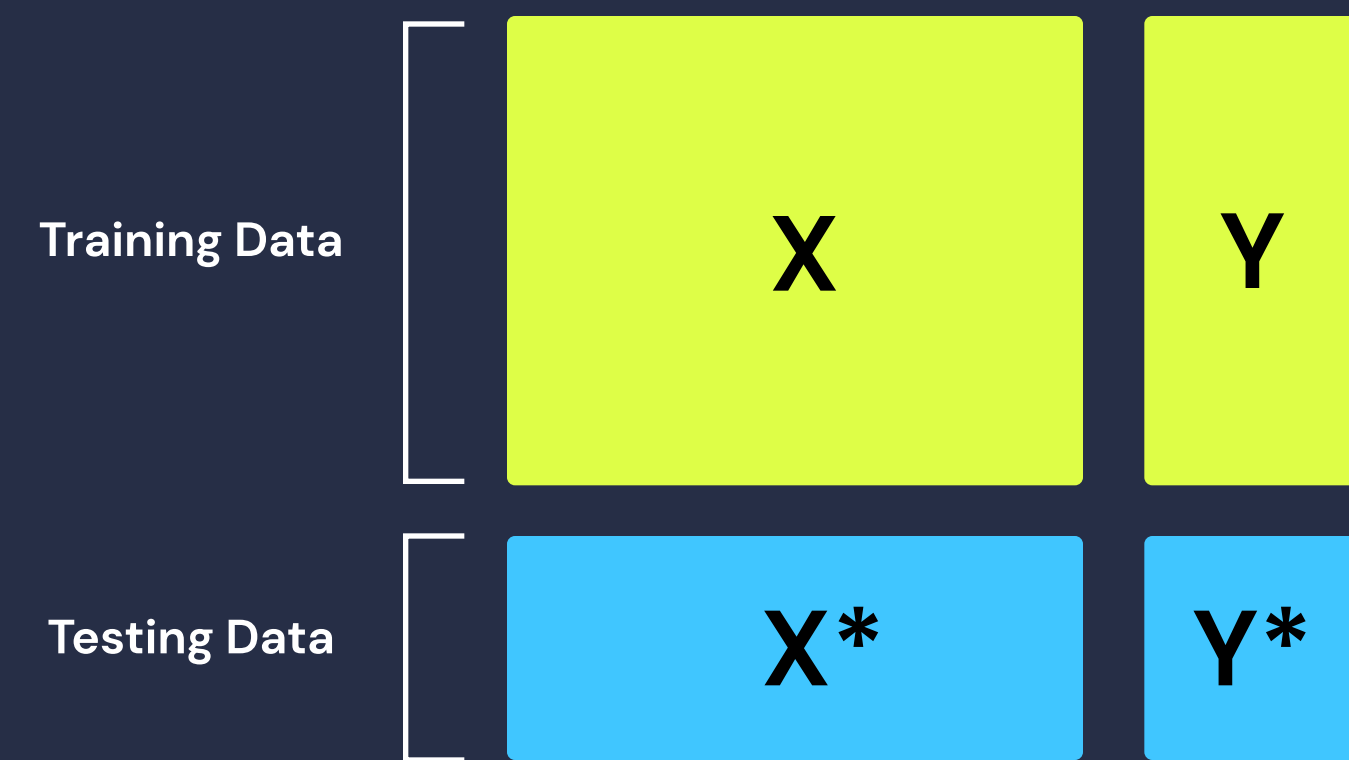
Iterative Refinement with Human Feedback

We refine the model by adding 10 rows of data at a time, each with the correct answer and the desired response format. However, there's a limitation with GPT 3.5 as it sometimes generates inaccurate predictions by adding extra information, like explanations, even when a simple format is preferred.

Positive

BERT & SEFIT - Active Learning

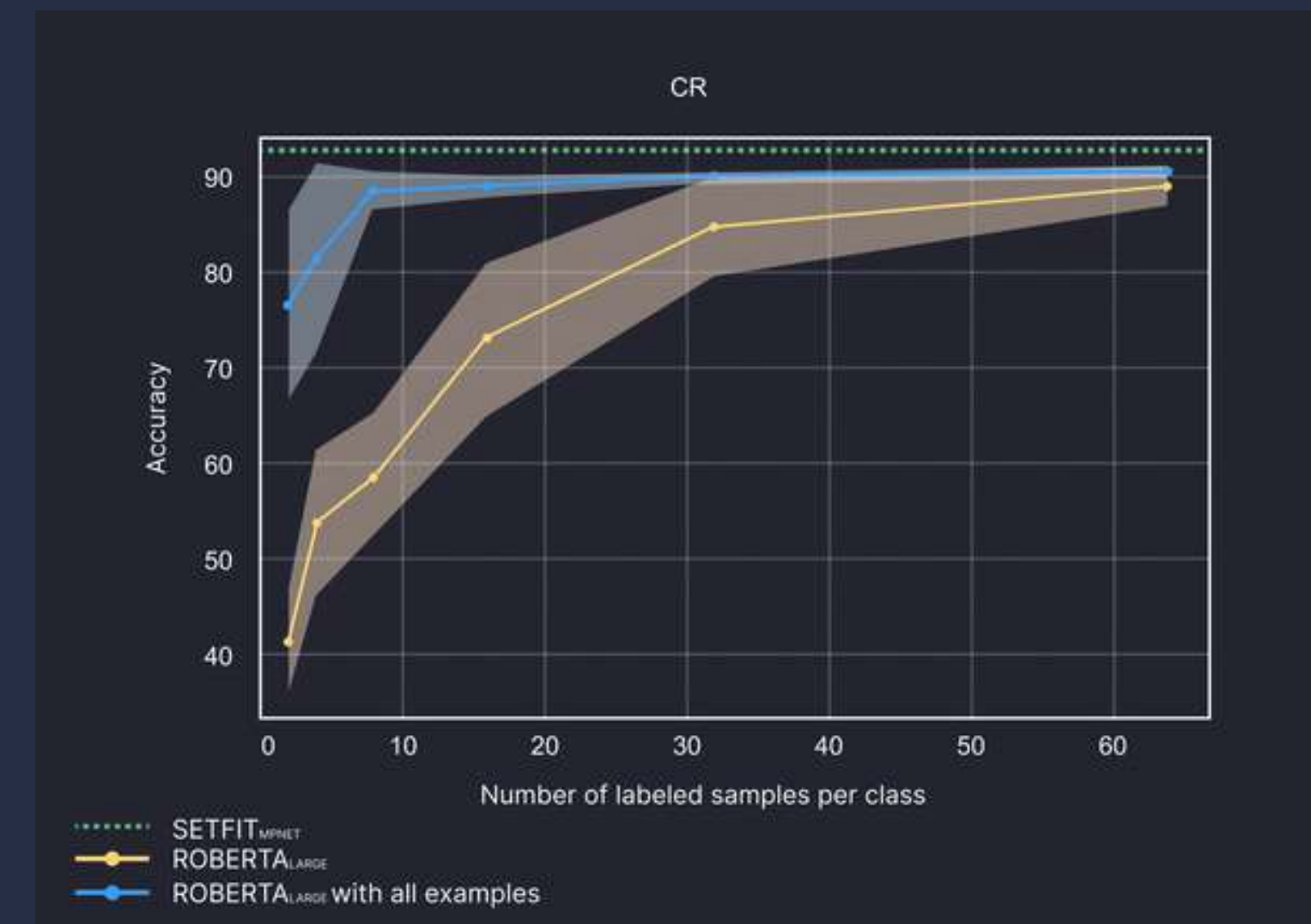
Traditional Machine Learning Models



Traditional AI approaches heavily depend on training models with extensive datasets containing millions of rows. This involves associating inputs, such as images or text, with accurate labels.

The primary challenge we face is twofold: How do we collect a substantial amount of labeled data efficiently, and how can we achieve this at a reduced cost?

Few Shot Learning



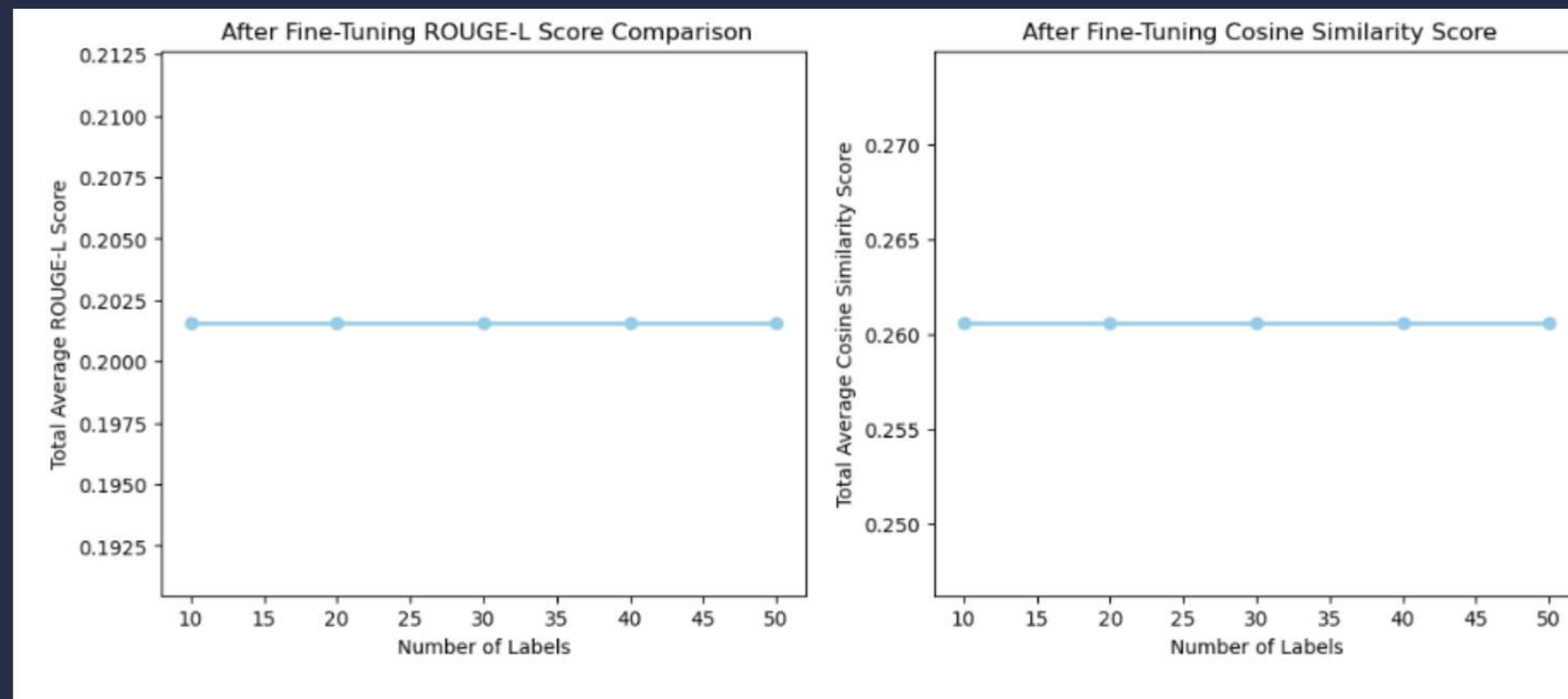
Few Shot Learning is where we provided model with few label examples to produce high accuracy

Feedback

The Market

Zero-shot Model Accuracy without Human Feedback

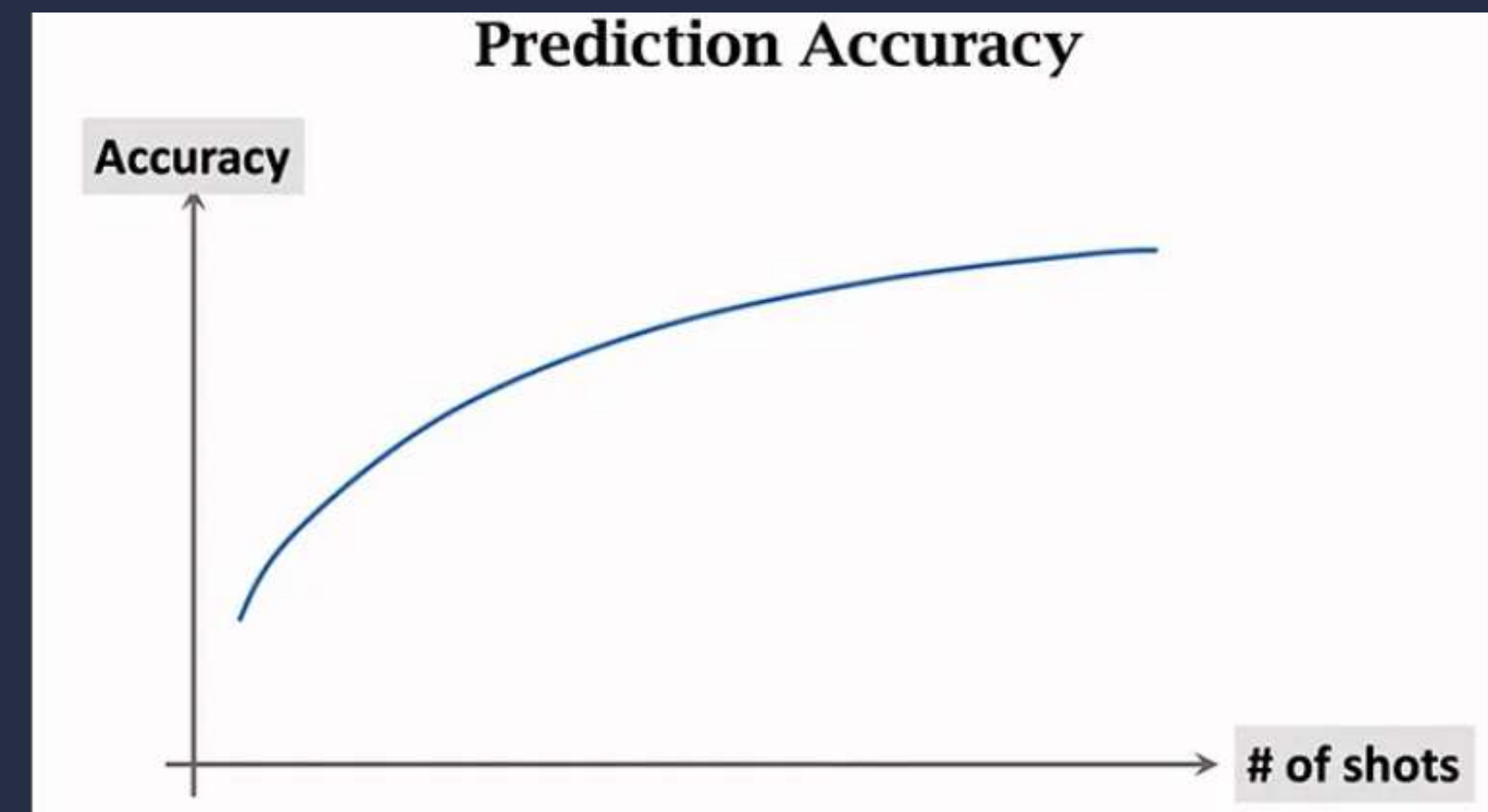
Without feedback, the model accuracy is horizontal and there is no sign of improvement.



Our Solution

With Human Feedback

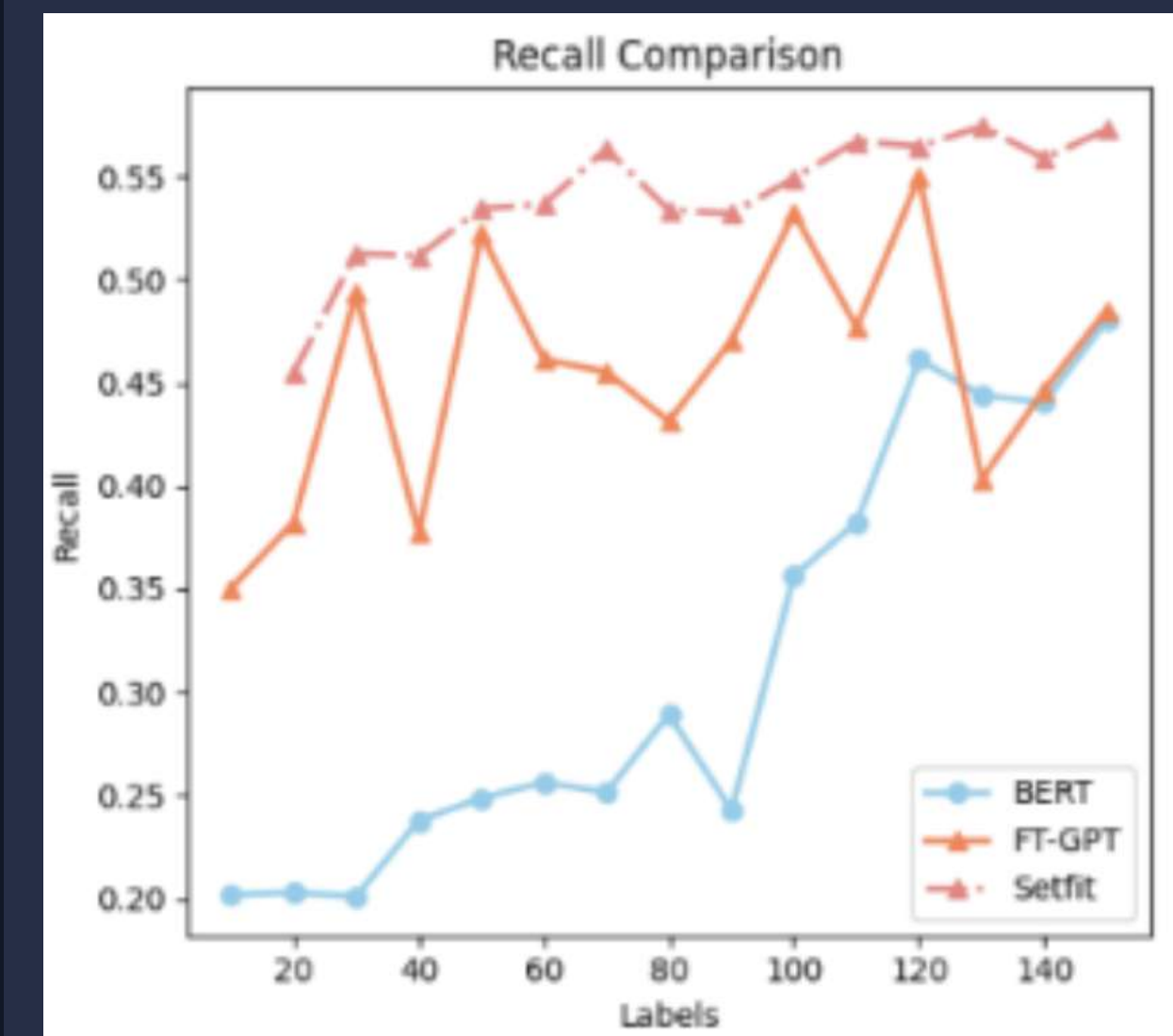
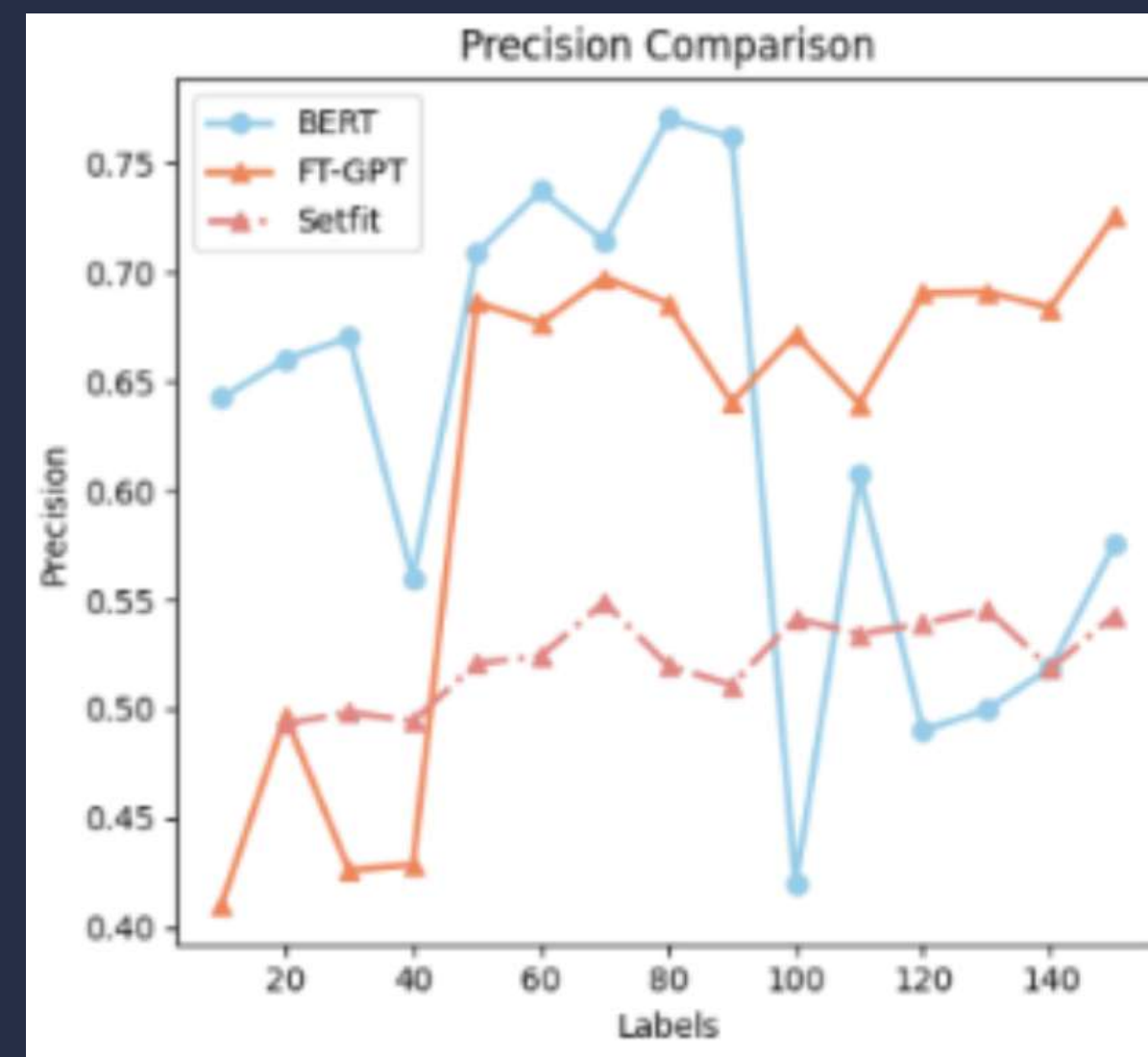
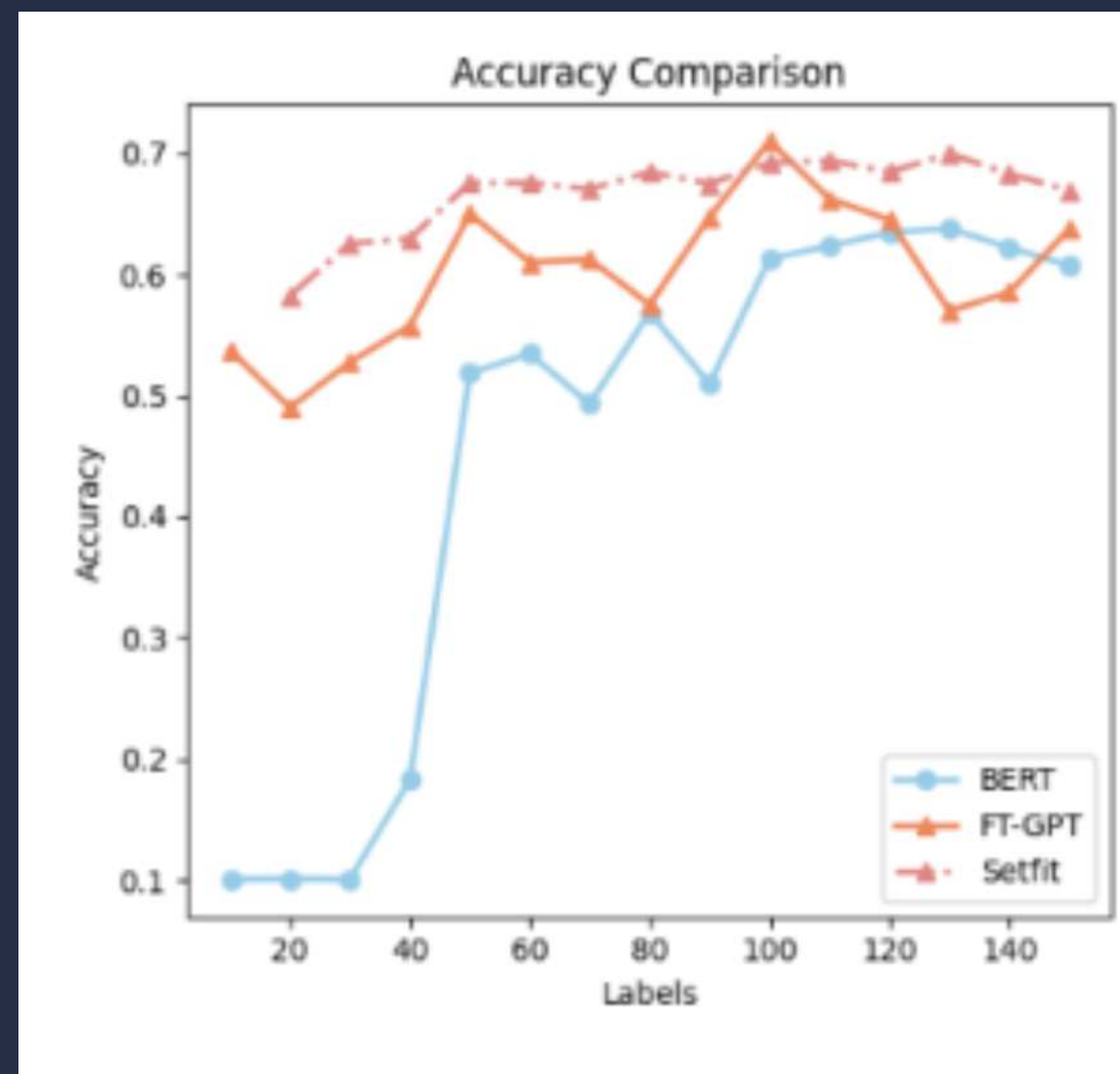
With human feedback, we see a growth in accuracy as we increase the amount of shots.



Fine-tuning Language Model + Human Feedback Performance

Figure 5: Amazon Dataset Plot

Performance Comparison on Amazon Dataset



Business Impact

Text Classification is crucial

for recognizing and managing risks, especially in industries where accurately organizing information is paramount (Healthcare, Finance).

Active learning contributes to cost savings

by streamlining processes and reducing errors.

Few shot learning

a component of active learning, enables equivalent model performance with fewer labeled data, reducing the resource-intensive nature of data labeling.

Implementing Active Learning enhances accuracy

but also optimizes resource utilization, making it a cost-effective solution.

Appendix

Potential Next Steps

- Ongoing benchmarking of accuracy
- Continue to refine and fine-tune these models with human feedback
- Further enhancements and optimizations in the future.

Summary

We learn that large language models will continually improve, and to enhance them, it's crucial to get better. We need to fine-tune these models with human feedback in order to achieve improvement, as you look at the timeline

Links

- [Improving Classification Performance With Human Feedback Research Paper](#)
- [The Code](#)